

D417

Network Automation and Deployment

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A regional ISP with 1,200 routers across three states is moving to an SDN design. The network team, mostly veterans of CLI work, wants one place to compute paths and push forwarding rules to every device. In an SDN architecture, where does the path-computing logic live?

- A. In a library of CLI scripts on each engineer's laptop, matching the team's experience.
- B. In a central controller that holds the control plane, separate from the devices' data plane.
- C. In the northbound business applications alone, with no controller needed.
- D. In each router's forwarding hardware (the data plane).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

An SDN controller uses OpenFlow to install flow entries on a set of switches. Which interface is this?

- A. The northbound interface.
- B. The data plane itself.
- C. The southbound interface.
- D. The east-west interface.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An engineer writes a playbook that makes sure VLAN 30 exists on 250 access switches. It runs nightly from a server in the Denver data center. On the second night, VLAN 30 already exists on every switch. What should a well-written playbook do?

- A. Always make at least one change, or the run counts as failed.
- B. Delete and re-create VLAN 30 on all 250 switches to guarantee a clean state.
- C. Report no changes and leave the switches as they are.
- D. Stop with an error because the VLAN already exists.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A script that runs an “add static route” command adds a duplicate route every time it runs. Which property is the script missing?

- A.** Scalability.
- B.** Encryption in transit.
- C.** More imperative, step-by-step commands.
- D.** Idempotency.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A junior engineer copies an inventory file from a vendor’s example, which was written for a 48-port switch model. The file is YAML. After editing, the automation tool can no longer parse it. The only change: two child keys under “interfaces” now start at the same column as “interfaces” itself. What is the cause?

- A.** Automation tools only ever read XML, so the file must be converted.
- B.** The file must list all 48 interfaces for a 48-port switch.
- C.** YAML requires curly braces around every object.
- D.** YAML uses indentation to show nesting, so the child keys are no longer under “interfaces.”

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A monitoring team pulls interface counters from a controller’s REST API every 60 seconds and shows them on a wall display in the NOC. Now an engineer needs to change only the description field of one interface without resending the whole interface object. Which HTTP method fits?

- A.** PATCH
- B.** GET
- C.** POST
- D.** PUT

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A script calling a controller's API gets back HTTP status 401. What does that mean?

- A.** The request lacked valid authentication credentials.
- B.** The controller had an internal fault.
- C.** The request succeeded but returned no body.
- D.** The requested resource does not exist.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A carrier's automation team wants a vendor-neutral, schema-defined way to read and change device configuration, with changes that roll back if any step fails. Their budget review is next quarter and the lead prefers Python. Which pairing fits?

- A.** Syslog forwarding to a central server.
- B.** NETCONF carrying YANG data models.
- C.** SNMP traps sent to a collector.
- D.** Python scripts that screen-scrape CLI output.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A mid-size hospital wants to automate 300 switches that cannot run third-party software. The CIO, who came from a software firm, favors tools with strong community support. Which approach fits the constraint?

- A.** An agent-based tool whose agent on each switch pulls its configuration.
- B.** An agentless tool that pushes changes over SSH or device APIs.
- C.** Whichever tool has the largest online community.
- D.** Any tool will do, since configuration tools never need software on devices.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

In an agent-based configuration model, how do managed nodes usually get their configuration?

- A.** An agent on each node checks in with a central server and pulls its desired state.
- B.** A control node pushes changes over SSH with nothing installed on the node.
- C.** Nodes share configuration with each other by broadcast.
- D.** Engineers paste configurations through the console.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A retail chain keeps its golden switch configurations in a Git repository. Last Friday, during a busy holiday sale, an engineer changed an ACL directly on a core switch from the CLI. The next pipeline run shows a difference. What is this, and what is the standard fix?

- A.** Treat the switch as the new source of truth and stop running the pipeline.
- B.** A merge conflict; delete the repository branch.
- C.** Leave it until the sale ends, because changes made during sales are exempt.
- D.** Configuration drift; route the change through the repository so Git stays the source of truth.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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12.

A script that configures 500 branch routers stores the admin password in plain text at the top of the file. The script is well commented and passes every test in the lab. What is the best fix?

- A.** Make the repository private, which fully secures the password.
- B.** Base64-encode the password inside the script.
- C.** Pull the credential from a secrets vault at run time and use a least-privilege service account.
- D.** Keep it as is, since the script passed lab testing.

Target word _____	The one fact that decides _____	Answer ____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A regional ISP ~~with 1,200 routers across three states~~ is moving to an SDN design. The network team, ~~mostly veterans of CLI work,~~ wants one place to compute paths and push forwarding rules to every device. In an SDN architecture, where does the path-computing logic live?

Target word: where control logic lives in SDN **Decides:** SDN separates the control plane into a central controller.

A	Camouflage	Answers the team's CLI background, which does not change where SDN puts control.
B	Correct	SDN's defining move is centralizing control decisions in a controller that programs the devices.
C	Dropped element	Applications request outcomes through the controller; removing the controller removes SDN's control plane.
D	Confusion set	The data plane forwards packets; it does not compute paths in SDN.

2. Answer C Concept 1 · transfer

An SDN controller uses OpenFlow to install flow entries on a set of switches. Which interface is this?

Target word: name the interface from the direction of traffic **Decides:** Controller-to-device communication is southbound.

A	Inverted operation	Northbound faces up, toward applications that ask the controller for outcomes.
B	Confusion set	The data plane is what gets programmed, not the interface that programs it.
C	Correct	OpenFlow carries instructions from the controller down to the devices.
D	Adjacent category	East-west links controllers to other controllers, not to switches.

3. Answer C Concept 2

An engineer writes a playbook that makes sure VLAN 30 exists on 250 access switches. It runs nightly ~~from a server in the Denver data center~~. On the second night, VLAN 30 already exists on every switch. What should a well-written playbook do?

Target word: behavior of a rerun at desired state **Decides:** The desired state is already met; an idempotent task changes nothing.

A	Absolute language	A run with zero changes is the expected result once the network is compliant.
B	Adjacent category	That is an imperative, redo-everything approach and causes needless disruption.
C	Correct	Idempotent automation compares actual to desired state and acts only on differences.
D	Confusion set	Existing state that matches the goal is success, not a conflict.

4. Answer D Concept 2 · transfer

A script that runs an “add static route” command adds a duplicate route every time it runs. Which property is the script missing?

Target word: name the missing property from the symptom **Decides:** Reruns change the result, so the task is not idempotent.

A	Adjacent category	Scale is about handling more devices, not repeat runs on one device.
B	Confusion set	Transport security does not stop duplicate entries.
C	Inverted operation	Step-by-step commands caused the problem; declaring the desired state fixes it.
D	Correct	An idempotent task would detect the route exists and do nothing.

5. Answer D Concept 3

A junior engineer copies an inventory file ~~from a vendor's example, which was written for a 48-port switch model.~~ The file is YAML. After editing, the automation tool can no longer parse it. The only change: two child keys under “interfaces” now start at the same column as “interfaces” itself. What is the cause?

Target word: why the file fails to parse **Decides:** YAML nesting is defined by indentation.

A	Absolute language	Most tools read YAML and JSON; the format was not the problem.
B	Camouflage	The port count does not affect whether the file parses.
C	Confusion set	Braces are JSON syntax; YAML relies on indentation.
D	Correct	Moving the keys to the parent's column changes the structure and breaks parsing.

6. Answer A Concept 4

A monitoring team pulls interface counters from a controller's REST API ~~every 60 seconds and shows them on a wall display in the NOC.~~ Now an engineer needs to change only the description field of one interface without resending the whole interface object. Which HTTP method fits?

Target word: method for a partial update **Decides:** Changing one field of an existing resource is a partial update.

A	Correct	PATCH applies a partial change to an existing resource.
B	Camouflage	Matches the counter-polling detail, but GET only reads.
C	Confusion set	POST creates a new resource.
D	Adjacent category	PUT replaces the whole resource; fields left out may be wiped.

7. Answer A Concept 4 · transfer

A script calling a controller's API gets back HTTP status 401. What does that mean?

Target word: meaning of a status code **Decides:** 401 is a client error: missing or invalid authentication.

A	Correct	401 Unauthorized means the caller has not proven who it is.
B	Adjacent category	Server faults are 5xx codes.
C	Inverted operation	Success codes are 2xx (204 for no content).
D	Confusion set	That is 404.

8. Answer B Concept 5

A carrier's automation team wants a vendor-neutral, schema-defined way to read and change device configuration, with changes that roll back if any step fails. ~~Their budget review is next quarter and the lead prefers Python.~~ Which pairing fits?

Target word: protocol and model for transactional config **Decides:** Schema-based plus rollback points to NETCONF with YANG.

A	Confusion set	Syslog records events; it does not configure devices.
B	Correct	YANG supplies the vendor-neutral schema; NETCONF supports candidate config, commit, and rollback.
C	Adjacent category	Traps are event notifications for monitoring, not transactional configuration.
D	Camouflage	Follows the Python preference but gives no schema and no rollback.

9. Answer B Concept 6

A mid-size hospital wants to automate 300 switches that cannot run third-party software. The CIO, ~~who came from a software firm, favors tools with strong community support.~~ Which approach fits the constraint?

Target word: tool model for devices that cannot host agents **Decides:** No software can be installed on the switches.

A	Adjacent category	Requires installing an agent, which the switches cannot run.
B	Correct	Agentless tools need nothing installed on the managed device.
C	Camouflage	Community size does not solve the no-agent constraint.
D	Absolute language	Agent-based tools do require software on each node.

10. Answer A Concept 6 • transfer

In an agent-based configuration model, how do managed nodes usually get their configuration?

Target word: direction of flow in agent-based tools **Decides:** Agents pull desired state on a schedule.

A	Correct	Pull-based check-ins are the agent model.
B	Inverted operation	That describes the agentless push model.
C	Confusion set	Configuration tools do not distribute state peer to peer.
D	Dropped element	Drops the automation entirely.

11. Answer D Concept 7

A retail chain keeps its golden switch configurations in a Git repository. Last Friday, ~~during a busy holiday sale,~~ an engineer changed an ACL directly on a core switch from the CLI. The next pipeline run shows a difference. What is this, and what is the standard fix?

Target word: name and remedy for out-of-band change **Decides:** The repo is the source of truth; the device no longer matches it.

A	Inverted operation	Reverses the IaC model and loses review and history.
B	Confusion set	Merge conflicts are between branches, not between the repo and a device.
C	Camouflage	The sale timing does not change what drift is or how it is fixed.
D	Correct	Either commit the ACL change through review or let the pipeline restore the approved state.

12. Answer C Concept 8

A script that configures 500 branch routers stores the admin password in plain text at the top of the file. ~~The script is well commented and passes every test in the lab.~~ What is the best fix?

Target word: secure handling of automation credentials **Decides:** A secret is hard-coded in the script.

A	Absolute language	Anyone with repo access, backups, or forks still sees the secret.
B	Confusion set	Encoding is reversible by anyone; it is not encryption.
C	Correct	Keeps secrets out of code and limits the damage if the account is misused.
D	Camouflage	Passing tests says nothing about how secrets are stored.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	11	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	SDN planes and interfaces	SDN moves control-plane decisions to a central controller; the controller talks down to devices over southbound APIs (OpenFlow, NETCONF) and up to applications over northbound APIs (usually REST).	Thinking forwarding hardware still computes paths, or swapping northbound and southbound.	“Controller,” “flow entries,” or an application asking the network for something.
2	Idempotency and declarative automation	An idempotent task describes the desired end state, so running it again changes nothing once that state is reached.	Believing automation must redo every step on every run.	“Runs nightly,” “second run,” or duplicate entries appearing after reruns.
3	Structured data formats	JSON uses braces and brackets; YAML shows nesting by indentation; XML uses tags. Tools parse structure, so a format error breaks the whole file.	Treating YAML indentation as cosmetic.	A file that fails to parse after a small edit.
4	REST methods and status codes	GET reads, POST creates, PUT replaces, PATCH partially updates, DELETE removes; 2xx = success, 4xx = client error (401 not authenticated, 403 not allowed, 404 not found), 5xx = server error.	Using PUT for a one-field change or reading 4xx as a server fault.	An API call that changes “only” one field, or a numeric response code.
5	Model-driven programmability	YANG defines the data model; NETCONF (XML over SSH) and RESTCONF (over HTTPS) carry it, with NETCONF supporting transactional commit and rollback.	Using SNMP or CLI scraping when the need is schema-based, transactional configuration.	“Vendor-neutral,” “schema,” “roll back if any step fails.”
6	Agent-based vs. agentless tools	Agentless tools (such as Ansible) push over SSH or APIs with nothing installed on the device; agent-based tools (such as Puppet or Chef) run an agent that pulls desired state from a server.	Picking an agent-based tool for devices that cannot run added software.	“Cannot install software” or “pulls its configuration.”
7	Infrastructure as code and drift	The version-controlled repository is the source of truth; a manual change on a device is drift, fixed by routing the change through the repo and pipeline.	Letting the live device become the new truth.	A pipeline showing a difference after a hand-made CLI change.

#	Concept	Rule	Trap	Tell
8	Securing automation	Keep secrets out of code (vault or injected at runtime), use least-privilege service accounts, and use encrypted transports (SSH, HTTPS), not Telnet.	Treating encoding or a private repository as protection for a plain-text password.	Credentials written inside a script or repo.

Course scope (public description): Network Automation and Deployment leverages previous experience in networking, scripting, and programming with the SDN paradigm. Students will create programs and scripts that automate network configuration across large networks. This course is designed as a hands-on experience where students will implement these techniques in a virtual space in order to produce a secure and functional deployed network.